

Final Report

ExpenseMate – Personal Expense Manager

Software Construction, Assignment 1 | Phase 6: Final Report & Demo | Release 1.1, 27 September 2026

1. Overview

ExpenseMate is a web app for tracking personal income and expenses. Users set monthly budgets, get warned before they overspend and see where their money goes in charts. It was built in six phases, from requirements to maintenance, and every phase was tested and measured.

144 / 144 automated tests pass	24 / 24 automated system tests pass on v1.1	13 / 13 defects found and fixed	94 ms slowest response with 10,000 transactions	1.99 average cyclomatic complexity
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2. Features

Feature	What it does
Accounts	Sign up, log in and log out. Each user sees only their own data.
Transactions	Add, edit, delete, filter and search income and expenses.
Multi-currency	Enter amounts in PKR, USD, EUR, GBP, AED or SAR; totals are shown in PKR.
Categories	Eight starter categories; add or remove your own.
Budgets and alerts	Monthly limit per category; warning at 80%, alert when over.
Analytics	Monthly summary, spending by category and a six-month income vs expense chart.
CSV import and export	Download a month as CSV; import a file, with bad rows listed by line number.

3. Architecture

Client–server at system level and **layered** inside the server (routes → controllers → services → repositories). Stack: React 19 and Vite on the client; Node.js 24, Express 5, Prisma 6 and SQLite on the server. Sessions use HttpOnly cookies and passwords are hashed with scrypt.

4. Work by Phase

Phase	Deliverable	Key result
1. Requirements	SRS (2 pages)	15 functional and 9 non-functional requirements; use case diagram
2. Architecture	Architecture document, Gantt chart	Style chosen and justified; logical, process, physical and deployment views
3. Construction	Source code, unit test report	3 modules (UI, Logic, DB); 91 unit tests; 95.6% logic coverage
4. Testing	Test cases, bug report, metrics	37 integration and 31 system tests; 12 defects logged; complexity measured
5. Maintenance	Change log, refactoring report	Multi-currency added; 2 bugs fixed; worst function 25 → 11; Lehman's laws applied
6. Final	This report, demo	Release 1.1: 3,753 lines of code in 73 files, 144 tests

5. Limitations and Future Work

- Exchange rates are fixed reference rates, not live market rates.
- SQLite suits one machine; cloud hosting would need PostgreSQL (a Prisma configuration change).
- Seven UI components are still above complexity 10 and should be split further.
- Portability was tested on Windows only; cloud backup (optional scope) was not built.

6. Lessons Learned

- **Each test level finds different bugs.** Unit tests missed every bug in routes and repositories; integration and system tests found them.
- **Measure, then refactor.** Complexity numbers showed exactly which function to fix first.
- **Small commits make changes safe.** Every change was small enough to test and undo on its own.
- **Changes create new bugs.** Adding login broke the phone navigation; only a system test caught it.

7. Demo Plan (about 10 minutes)

#	Show	Time
1	Run the app (commands below); open http://localhost:5173 and sign up a new user	1 min
2	Starter categories; add an expense with N , then an income in USD with the $\approx$ Rs preview	2 min
3	Set a Food budget; add spending until it turns to <i>Close to limit</i> , then <i>Over budget</i>	2 min
4	Overview page: summary, category chart and six-month trend	1 min
5	Export September as CSV; import a file with one bad row to show the line-number error	2 min
6	Log out; log in as the demo account to show that data is private per user	1 min
7	Run the tests: <code>npm test</code> in server and client (144 passing)	1 min

To run: in *server*: `npm install, npm run db:reset, npm run dev`. In *client*: `npm install, npm run dev`. The demo account can be opened with one click on the login page.

8. Submission Bundle

Item	File
SRS document	docs/Phase1_SRS_ExpenseMate.pdf
Architecture diagrams	docs/Phase2_Architecture_ExpenseMate.pdf
Source code (ZIP)	Project repository (client, server, docs)
Test cases and results	docs/Phase3_Construction_UnitTests_ExpenseMate.pdf, docs/Phase4_Integration_SystemTesting_ExpenseMate.pdf
Metrics report	docs/Phase4_Integration_SystemTesting_ExpenseMate.pdf (sections 5–6), docs/Phase5_Maintenance_ExpenseMate.pdf
Final report	docs/Phase6_Final_Report_ExpenseMate.pdf